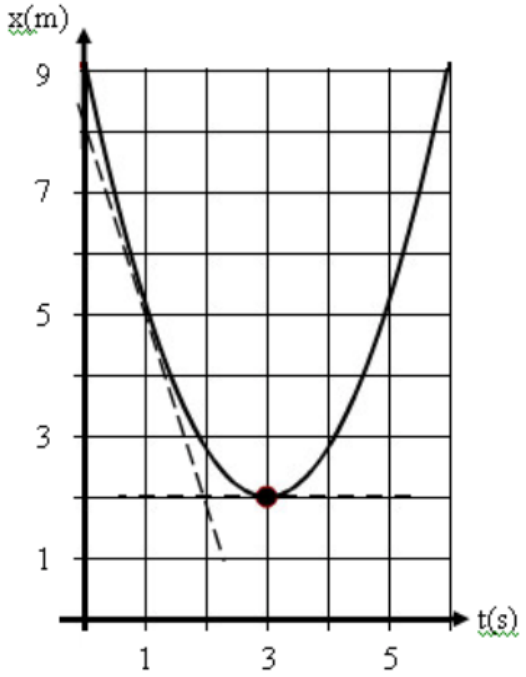
The position-time curve for a particle moving along the x-axis is shown in **Figure** . The dashed straight lines are tangent to the curve at t = 1.0 s and t = 3.0 s. Find the magnitude of the average acceleration in the time interval t = 1.0 s and t = 3.0 s.

Fig#



A) 1.5 m/s^2
B) 5.0 m/s^2
C) 1.0 m/s^2
D) 2.0 m/s^2
E) 2.5 m/s^2

The position of a particle as a function of time is given by $x(t) = 3.0t^3 - 10.0t^2 + 9.0t$, where x is in meters and t is in seconds. Find the velocity at the time when the acceleration is zero.

A) – 2.1 m/s
B) + 2.1 m/s
C) + 5.0 m/s
D) – 5.0 m/s
E) + 3.5 m/s

The velocity of a particle is given by $v = At^2 + (B/A)\,t$, where v is in m/s and t is in seconds. The dimension of B is:

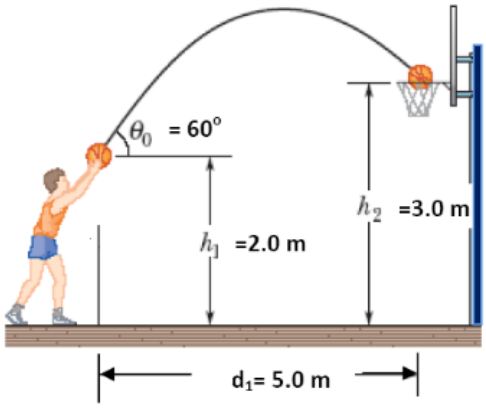
A) $L^2\,T^{-5}$
B) $L^3\,T^{-3}$
C) $L^2\,T^3$
D) $L^4\,T^4$
E) $L^5\,T^{-6}$

A player running in an open field has components of velocity $v_x = 2.60$ m/s and $v_y = -1.80$ m/s at t = 10.0 s. For the time interval from t = 10.0 s to t = 20.0 s, the magnitude of its average acceleration is 0.45 m/s^2 and makes an angle of 31.0° with the positive x-axis. Find the velocity (in units m/s) of the player at t = 20.0 s.

A) 6.46 **i** + 0.518 **j**
B) 5.32 **i** + 3.52 **j**
C) 4.52 **i** + 5.60 **j**
D) 2.46 **i** + 2.52 **j**
E) 9.60 **i** + 5.30 **j**

A basketball player would like to throw a ball at an angle of $\theta_0 = 60°$ above the horizontal such that the ball just goes through the center of the rim of the basket that is $h_2 = 3.0$ m high from the floor and it is at a horizontal distance of $d_1 = 5.0$ m from the player's hand (see **Figure**). At the instant the ball leaves the player's hand, his hand is $h_1 = 2.0$ m above the floor. Find the magnitude of the initial velocity of the ball.

Fig#



A) 8.0 m/s
B) 5.0 m/s
C) 9.8 m/s
D) 3.2 m/s
E) 7.0 m/s

A ball starts from rest and slides down a hill with a constant acceleration for 10.0 s. If it travels 50.0 m during the first 5.00 seconds of its motion, how far will it travel during the next 5.00 seconds of its motion?

A) 150 m
B) 98.0 m
C) 75.0 m
D) 200 m
E) 25.0 m

If vector **A** = b (3.0 **i** + 4.0 **j**), where b is a constant. Find the value of b that makes vector **A** a unit vector.

A) 0.20
B) 0.10
C) 0.30
D) 0.50
E) 0.40

Two vectors $\vec{A}$ and $\vec{B}$ have magnitudes 3.0 and 4.0 respectively. Their vector product is $\vec{A}\times\vec{B} = -5.0\vec{k} + 2.0\vec{i}$. Find the angle between $\vec{A}$ and $\vec{B}$.

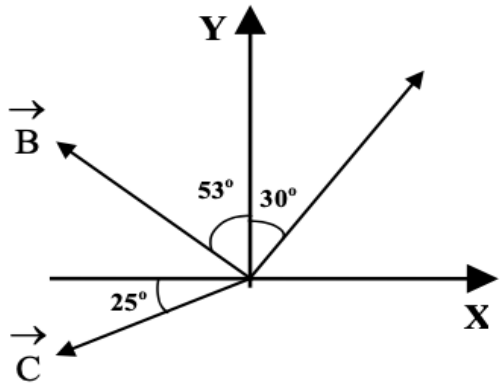
A) 27°
B) 22°
C) 39°
D) 11°
E) 16°

Two objects A and B are thrown from the top of a building with the same magnitude of velocity. Object A is thrown upward but object B is thrown downward. When they reach the ground (ignore air resistance):

A) The two objects have the same velocity.
B) Object A has a higher velocity than object B.
C) Object B has a higher velocity than object A.
D) The velocities will depend on the masses of the objects.
E) The velocities will depend on the shapes of the objects.

Figure shows three vectors **A**, **B** and **C**, whose magnitudes are 15 m, 10 m and 12 m respectively. Find both the magnitude and direction of the resultant of theses vectors.

Fig#



A) 18.0 m, 129° from the positive x-axis
B) 14.0 m, 150° from the positive x-axis
C) 12.0 m, 53° from the positive x-axis
D) 20.0 m, 163° from the positive x-axis
E) 10.0 m, 110° from the positive x-axis